

NO.	ANSWER SCHEME
1.	$y = \frac{m\lambda D}{d}$ $= \frac{(1)(546 \times 10^{-9})(60 \times 10^{-2})}{0.12 \times 10^{-3}}$ $= 2.73 \times 10^{-3} m$
2.	$d = 0.05 \times 10^{-3}$ $D = 0.75 m$ $y = 12 \times 10^{-2} m$ $m = 7$ $y = \frac{m\lambda D}{d}$ $12 \times 10^{-2} = \frac{(7)\lambda(0.75)}{0.05 \times 10^{-3}}$ $\lambda = 1.14 \times 10^{-6} m$
3	(a) $y_m = \frac{m\lambda D}{d}$ $4.0 \times 10^{-3} = \frac{(2)(\lambda)(1)}{0.3 \times 10^{-3}}$ $\lambda = 6.0 \times 10^{-7} m$ (b) $x_{\left(m+\frac{1}{2}\right)} = \frac{\left(m+\frac{1}{2}\right)\lambda D}{d}$ $x_{\left(2+\frac{1}{2}\right)} = \frac{\left(2+\frac{1}{2}\right)6.0 \times 10^{-7}(1)}{0.3 \times 10^{-3}}$ $= 5.0 \times 10^{-3} m$
4.	$\Delta y = \frac{\lambda D}{d}$ $6.25 \times 10^{-4} = \frac{(6.2 \times 10^{-7})(0.8)}{d}$ $d = 7.94 \times 10^{-4} m$
5.	(a) (b)

	$\Delta y = \frac{\lambda D}{d}$ $= \frac{(7.0 \times 10^{-7})(0.4)}{0.2 \times 10^{-3}}$ $= 1.4 \times 10^{-3} m$	$\Delta y = \frac{\lambda D}{d}$ $= \frac{(4.2 \times 10^{-7})(0.4)}{0.2 \times 10^{-3}}$ $= 0.84 \times 10^{-3} m$
6.	(a) $y = \frac{m\lambda D}{d}$ $= \frac{(3)(600 \times 10^{-9})(1)}{0.24 \times 10^{-3}}$ $= 7.5 \times 10^{-3} m$	(b) $x_{\left(m+\frac{1}{2}\right)} = \frac{\left(m+\frac{1}{2}\right)\lambda D}{d}$ $x_{\left(2+\frac{1}{2}\right)} = \frac{\left(2+\frac{1}{2}\right)600 \times 10^{-9}(1)}{0.24 \times 10^{-3}}$ $= 6.25 \times 10^{-3} m$
7.	$y_m = \frac{m\lambda D}{d}$ $7.5 \times 10^{-3} = \frac{(5)(\lambda)(1)}{0.2 \times 10^{-3}}$ $\lambda = 3.0 \times 10^{-7} m$	
8.	$\Delta y = \frac{4.5 \times 10^{-3}}{5} = 0.9 \times 10^{-3} m \quad \frac{D}{d} = 1500$ $\Delta y = \frac{\lambda D}{d}$ $0.9 \times 10^{-3} = \lambda(1500)$ $\lambda = 6.0 \times 10^{-7} m$	
9.		

	$x_{\left(m+\frac{1}{2}\right)} = \frac{\left(m+\frac{1}{2}\right)\lambda D}{d}$ $2.03\times 10^{-2} = \frac{\left(4+\frac{1}{2}\right)\lambda(1.5)}{0.2\times 10^{-3}}$ $\lambda = 6.01\times 10^{-7} \text{ m}$
10.	$\Delta y = x_3 - y_2$ $= \frac{\left(3+\frac{1}{2}\right)600\times 10^{-9}(1)}{0.3\times 10^{-3}} - \frac{(2)600\times 10^{-9}(1)}{0.3\times 10^{-3}}$ $= 7.0\times 10^{-3} - 4\times 10^{-3}$ $= 3\times 10^{-3} \text{ m}$